

Classification Model Evaluation:
Binary Classification of Wine Quality and Multiclass Classification of Breast Cancer Subtype
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Abstract

This project evaluates supervised classification models on two real-world datasets: a binary wine quality dataset and a multiclass breast cancer gene expression dataset. The wine dataset merges red and white wines from the UCI Wine Quality collection and aims to predict human quality ratings as a high vs low binary classification problem, modeled using Logistic Regression, Support Vector Machine (SVM) with an RBF kernel, Decision Tree, and Random Forest with stratified cross-validation and accuracy, F1-score, and ROC-AUC as key evaluation metrics. The breast cancer dataset consists of 151 samples with 54,675 gene expression features and six molecular subtypes, and is modeled using Kernel SVM, Random Forest, Multinomial Logistic Regression, and Naive Bayes with macro F1 and balanced accuracy as the primary metrics. Hyperparameter tuning and learning-curve analysis reveal that Random Forest offers the best trade-off between minority-class F1 and ROC-AUC for wine quality, while regularized Logistic Regression with feature selection achieves perfect test performance on breast cancer subtypes, highlighting the strength of linear models in high-dimensional genomics.

Introduction

The purpose of this project is to deepen understanding of classification problems through the application of machine learning algorithms to two distinct real-world datasets. Binary classification assigns data to one of two possible classes, while multiclass classification assigns data to one of three or more possible classes. Each problem type presents unique challenges that require careful model selection, preprocessing, and evaluation strategy.

The binary classification problem involves predicting wine quality (high vs low/average) using a merged red and white wine dataset derived from the UCI Wine Quality study. The inputs are physicochemical measurements such as acidity, residual sugar, sulfur dioxide levels, and alcohol content, and the goal is to identify which wines fall into the higher quality range. This task is practically relevant for winemakers, distributors, and quality control settings, where being able to flag potentially high-quality wines based only on lab measurements could support better inventory and pricing decisions.

For the binary wine quality task, four algorithms are used: Logistic Regression, Support Vector Machine (SVM) with an RBF kernel, Decision Tree, and Random Forest. Logistic Regression provides a linear baseline that is easy to interpret and works well when features are scaled and regularized, it helps show how far simple linear decision boundaries can go. The RBF-kernel SVM is included to capture nonlinear relationships between the physicochemical features and wine quality that a linear model might miss, while still supporting class-weighting to address imbalance. The Decision Tree offers a transparent, rule-based model that directly exposes which feature thresholds (for example, specific ranges of alcohol or sulphates) separate high-quality from low/average wines. Random Forest, an ensemble of many decision trees, reduces the high variance of a single tree and is well suited to modeling complex interactions in tabular data, typically providing stronger generalization than a single tree on moderately imbalanced datasets.

The multiclass classification problem involves predicting breast cancer molecular subtypes from gene expression data. Breast cancer is not a single disease but a collection of molecularly distinct subtypes including Basal, HER, Luminal A, Luminal B, Cell Line, and Normal, each with different prognoses and treatment

responses. Accurate subtype classification from gene expression profiles could support clinical decision-making and personalized medicine. This problem is particularly challenging because the dataset has 54,675 gene features relative to only 151 samples, creating a high-dimensional, low-sample-count regime that is prone to overfitting.

For the breast cancer dataset, the following four algorithms were implemented: Kernel SVM with an RBF kernel, Random Forest, Multinomial Logistic Regression, and Naive Bayes. Each algorithm was chosen for specific reasons tied to the characteristics of this dataset. Kernel SVM was selected because it finds an optimal maximum-margin hyperplane in a transformed feature space, which is well suited for settings where the number of features greatly exceeds the number of samples. By mapping data into a higher-dimensional space using the RBF kernel, SVM can capture non-linear relationships between gene expression patterns and cancer subtypes without being overwhelmed by the raw dimensionality. Random Forest was chosen because it naturally handles high-dimensional data by randomly subsampling features at each split, effectively performing built-in feature selection. Averaging hundreds of decision trees reduces variance and produces stable predictions even when individual trees overfit. Logistic Regression was selected as a linear baseline because L1 and L2 regularization directly penalize large coefficients, forcing the model to rely on a smaller subset of the most informative genes rather than trying to fit all 54,000 features. This embedded feature selection behavior is particularly well suited for genomics where most genes are irrelevant to any specific classification task. Naive Bayes was chosen because it is computationally efficient in high-dimensional spaces and assumes feature independence, an assumption that is technically violated in gene expression data due to biological co-regulation but holds well enough in practice for classification. KNN was explicitly excluded because in high-dimensional spaces all pairwise distances become nearly equal, making neighborhood-based reasoning unreliable. This project depends on data preprocessing, hyperparameter tuning, and performance metric evaluation to determine which algorithms are most effective for each dataset. The analysis compares algorithm performance at the inter-model level under varying hyperparameter conditions, and challenges encountered are discussed alongside corrective measures.

Datasets

The binary dataset, “Wine Quality — Red & White,” was obtained from Kaggle and is based on the original UCI Wine Quality study of Portuguese red and white wines, where researchers collected physicochemical measurements and human sensory ratings. The target variable is the quality score, an ordinal rating on a 0–10 scale, for this project, these scores are converted into a binary label, `quality_binary`, where scores from 7 to 9 indicate high-quality wines and scores from 3 to 6 indicate low/average wines. After merging and cleaning, the dataset contains 5,320 unique wine samples with 12 predictor variables: 11 continuous physicochemical measurements (such as acidity, residual sugar, density, pH, sulfur dioxide, sulphates, and alcohol) and one categorical wine type indicator encoded as a binary feature `type_bool` (0 for red, 1 for white).

All predictors are numeric after encoding, and there are no missing values in the features or the target. Approximately 18% of the original rows are exact duplicates, representing repeated measurements of the same wine; these duplicates are removed so that each remaining row corresponds to a distinct sample. The original quality scores are moderately imbalanced, with most wines rated 5 or 6, and after binarization the high-quality class remains the minority. The data are split into 80% training and 20% testing using a stratified split with a fixed random seed so that the high-quality class is proportionally represented in both sets. Key challenges for this dataset include moderate class imbalance, strong correlations among several physicochemical features, and overlapping feature distributions between mid-quality and high-quality wines that make the decision boundary fuzzy.

The multiclass dataset is the Breast Cancer Gene Expression Dataset (CuMiDa BRCA Subtypes), obtained from Kaggle and compiled by Bruno Iochins Grisci as part of the CuMiDa project, a curated microarray database for machine learning research. Gene expression measurements were collected using

microarray technology to quantify mRNA expression levels in breast cancer tumor samples. The original purpose of the dataset was to analyze gene expression patterns across different molecular subtypes of breast cancer and to support cancer subtype classification research.

The target variable is the column "type," which represents the molecular subtype of each patient sample, making this a six-class multiclass classification problem. The six target classes are: Basal (41 samples), an aggressive subtype associated with triple-negative breast cancer; HER (30 samples), characterized by overexpression of the HER2 gene; Luminal B (30 samples), a hormone receptor-positive subtype with higher proliferation rates; Luminal A (29 samples), the most common and least aggressive hormone receptor-positive subtype; Cell Line (14 samples), lab-grown cancer cell samples used for research purposes; and Normal (7 samples), non-cancerous breast tissue. Correctly classifying each subtype is clinically critical because each one requires a different treatment approach. The dataset contains 151 samples and 54,675 features, each representing the expression intensity of a specific gene probe such as 1007_s_at and 1053_at. All features are continuous numerical variables. There are no categorical predictor variables, no missing values, and no duplicate samples. This makes the dataset a high-dimensional, low-sample-count problem where the number of features far exceeds the number of samples ($d \gg n$), with a feature-to-sample ratio of approximately 362 to 1. The dataset is moderately imbalanced, with the Basal class containing 41 samples and the Normal class containing only 7. The data was split into 70% training and 30% testing using stratified sampling with `random_state=42`, yielding 105 training samples and 46 test samples. Stratification ensured that all six classes were proportionally represented in both sets. The primary challenges include extreme dimensionality relative to sample size, which increases the risk of overfitting and the curse of dimensionality, as well as class imbalance that complicates reliable evaluation of minority class performance.

Methods

Data Processing

For the wine quality dataset, preprocessing is performed once at the start and the cleaned dataset is reused for all steps. The original wine type column is encoded as a numeric feature `type_bool`, mapping red wines to 0 and white wines to 1, and the original type column is then removed. The integer quality scores are transformed into the binary target, `quality_binary`, where scores 7–9 represent high-quality wines and scores 3–6 represent low or average wines, and the original quality column is removed. Exact duplicate rows are removed and the index is reset so that repeated measurements of the same wine do not bias model training. After these steps, the dataset consists entirely of numeric predictor variables and a binary target.

StandardScaler normalization is applied for models that are sensitive to feature scale. Specifically, Logistic Regression and SVM are implemented in pipelines that first apply scaling and then train the classifier, ensuring that features measured in different units (such as acidity, sulfur dioxide, and alcohol content) contribute comparably to the loss function. Decision Tree and Random Forest are trained without scaling, because tree-based methods partition the feature space using thresholds and are invariant to monotonic rescaling of individual features. Class imbalance is handled in two ways: all classifiers are configured with `class_weight="balanced"` so that misclassifying the minority high-quality class is penalized more heavily, and the Logistic Regression pipeline also includes a `RandomUnderSampler` to reduce the majority class before fitting.

A three-step feature reduction pipeline was used to reduce the 54,000+ gene features and limit overfitting. First, 62 AFFX control probes were removed since they are technical markers with no biological relevance. Second, variance filtering (applied after the train/test split) removed genes with little variation across samples, as they provide no useful information for classification. Third, `SelectKBest` was applied to Logistic Regression and Naive Bayes to retain the most informative genes ($K = 100\text{--}500$), with selection performed

inside the cross-validation pipeline to prevent data leakage. Random Forest did not use this step due to built-in feature selection, while Kernel SVM used PCA to reduce dimensionality for efficient computation.

Partitioning and Metrics

For the binary wine quality task, the cleaned dataset is partitioned into 80% training and 20% testing using a stratified split on `quality_binary` with a fixed random seed, preserving the class proportions in both subsets. Within the training set, stratified 7-fold cross-validation is used for both performance estimation and hyperparameter tuning, providing stable evaluation while maintaining representative samples of the minority high-quality class in each fold. Several metrics are used to capture performance under moderate class imbalance. Accuracy is reported as an overall measure of correctness, but it can be misleading because predicting only the majority class can still yield high accuracy. To address this, F1-score is computed with the high-quality wines treated as the positive class, this provides a single score that balances precision and recall. ROC-AUC is also reported to evaluate how well each model separates high-quality from low-quality wines across all decision thresholds. F1 has the disadvantage of summarizing performance at a single point on the precision–recall curve, and ROC-AUC can remain high even if the operating point is suboptimal, but together with accuracy these metrics provide a more complete picture of performance on the minority class than any single measure alone.

For the multiclass breast cancer dataset, stratified 5-fold cross-validation was used as the primary evaluation method because 151 total samples is too few for a single train/test split to provide reliable performance estimates. Stratification ensured that all six classes, including the minority Normal class with only 7 samples, were proportionally represented in every fold. A value of $k=5$ was chosen over $k=10$ because $k=10$ would produce folds too small for reliable estimates given the minority class sizes. Two metrics were chosen for evaluation: Macro-Averaged F1-Score and Balanced Accuracy. Both were selected specifically because of the class imbalance present in the dataset. Plain accuracy was avoided because the six classes are very unequal in size, with Basal having 41 samples and Normal having only 7. A model that completely ignores the Normal and Cell Line classes could still score approximately 85% plain accuracy by correctly predicting only the majority classes, which would be a clinically dangerous result since misclassifying a cancer subtype directly affects treatment decisions. Balanced Accuracy computes recall separately for each of the six classes and averages those values equally, giving every subtype the same weight regardless of how many samples it has. A balanced accuracy of 0.17 represents random chance among six classes, meaning any score well above this threshold indicates genuine learning. A score of 1.0 means the model correctly identified every sample in every class including the smallest minority classes. This metric directly aligns with the clinical goal of reliably classifying all subtypes regardless of rarity. Its disadvantage is that it ignores precision, meaning a model that generates many false positives for a given class could still appear to perform well. Macro F1-Score addresses this by computing F1 independently for each class and averaging equally, balancing both precision and recall per subtype. This forces the model to perform well on all six classes simultaneously rather than trading off one for another. Its disadvantage is that it treats all classes as equally clinically important even when some subtypes such as Basal may carry higher urgency due to their aggressive nature. Together these two metrics provide a complete and clinically grounded picture of model performance under class imbalance, one that plain accuracy alone could not deliver.

Hyperparameter Tuning

For the wine quality dataset, hyperparameter tuning is performed for all four models using grid search with stratified 7-fold cross-validation on the training data. The optimization metric is the F1-score for the high-quality class to account for class imbalance. For Logistic Regression, undersampling and feature scaling are applied, tuning the regularization strength C and `l1_ratio` to control model complexity and sparsity. For the RBF-kernel SVM, C and γ (gamma) are tuned to balance margin width and decision boundary flexibility. For the Decision Tree, `max_depth`, `min_samples_leaf`, and `min_samples_split` are tuned to control tree complexity

and prevent overfitting. For the Random Forest, `n_estimators`, `max_depth`, and `max_features` are tuned to adjust ensemble size, tree complexity, and feature sampling.

For the breast cancer dataset, Grid Search with stratified 5-fold cross-validation was used for Kernel SVM and Logistic Regression where the search space was small enough to evaluate exhaustively. The SVM grid searched `C` values of 0.1, 1, and 10, and `gamma` values of 0.001, 0.01, and scale. The Logistic Regression grid searched `C` values of 0.01, 0.1, 1, and 10, and penalty types of L1 and L2. Randomized Search was used for Random Forest and Naive Bayes where exhaustive search would be computationally expensive. Random Forest searched `n_estimators` values of 100, 200, and 300, and `max_features` values of `sqrt` and `log2`. Naive Bayes searched `var_smoothing` values ranging from $1e-9$ to $1e-3$ and `SelectKBest` `K` values of 100, 200, and 300. All tuning was scored using balanced accuracy. Class weighting (`class_weight='balanced'`) was applied to SVM and Logistic Regression to prevent those models from ignoring minority classes during training.

Hyperparameter Tuning and Other Analyses

Binary Classification of Wine Quality

[Logistic Regression]

The tuning curve for Logistic Regression shows that training and validation F1-scores remain very similar across all tested values of `C`, with only minimal variation. This indicates stable behavior and effective regularization without overfitting, as performance is not highly sensitive to the choice of `C`. The learning curve shows that validation performance increases slightly as more training examples are added, while the training and validation curves remain close together. This suggests low variance and consistent generalization across folds. The relatively small and narrowing gap between the curves indicates that the model remains stable as dataset size increases, with limited benefit from additional data.

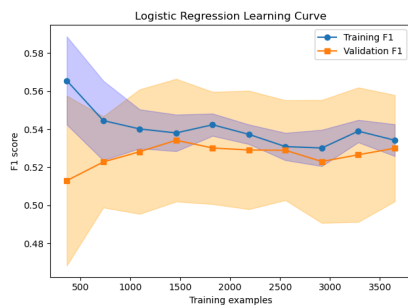


Fig. 1A. Logistic Regression
Learning Curve (F1-score vs. Training Set Size)

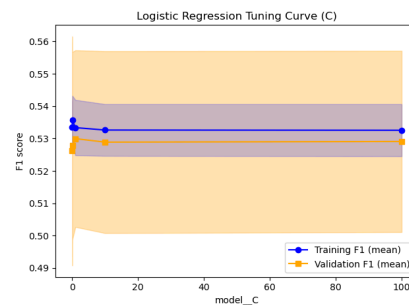


Fig. 1B. Logistic Regression Hyperparameter Tuning Curve
(F1-score vs. Regularization Strength CCC)

[SVM (RBF)]

The tuning curves show that the SVM is sensitive to hyperparameters. As `C` and γ increase, training F1-score rises, but validation F1 remains relatively flat, indicating overfitting. The best performance occurs at moderate parameter settings. The learning curve shows a consistent gap between training and validation F1-scores, suggesting higher variance than Logistic Regression. As more training data is added, validation performance improves gradually and the gap narrows slightly, but does not fully close. Overall, the SVM performs competitively but remains sensitive to hyperparameter tuning and exhibits higher variance.

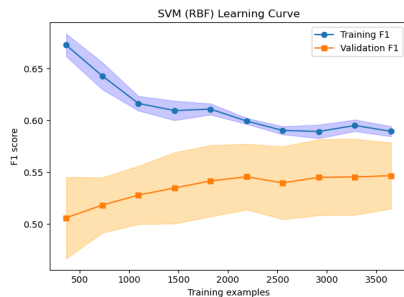


Fig. 2A. SVM (RBF)
Learning Curve
(F1-score vs. Training Set Size)

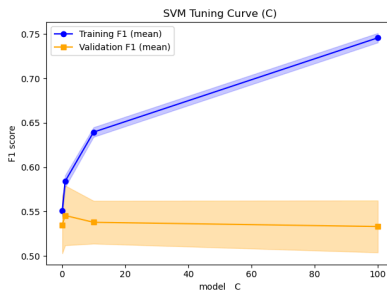


Fig. 2B. SVM (RBF) Hyperparameter
Tuning Curve (F1-score vs.
Regularization Parameter CCC)

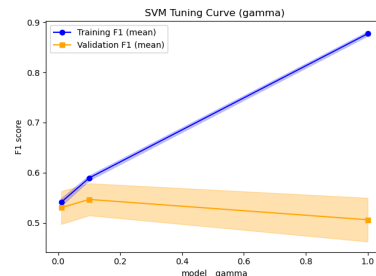


Fig. 2C. SVM (RBF) Hyperparameter
Tuning Curve (F1-score vs. Kernel
Parameter γ (gamma))

[Decision Tree]

The tuning curve for max depth shows clear overfitting behavior. As depth increases, the training F1-score rises sharply, while the validation F1-score peaks at a moderate depth and then declines. This indicates that deeper trees memorize the training data but generalize poorly. The learning curve also reflects high variance. The training F1-score starts relatively high and decreases slightly as more data is added, while the validation F1-score improves but remains noticeably lower. The gap between the curves indicates overfitting, although additional data helps reduce it slightly. Overall, the Decision Tree exhibits high variance and weaker generalization compared to other models, especially at higher depths.

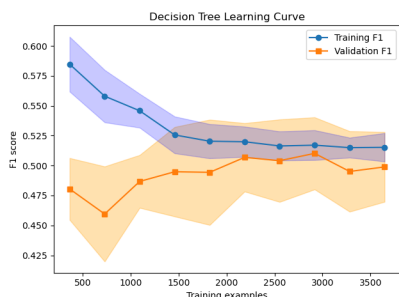


Fig. 3A. Decision Tree Learning Curve (F1-score vs.
Training Set Size)

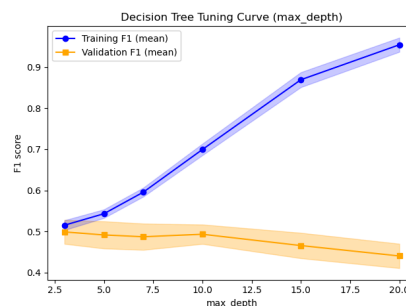


Fig. 3B. Decision Tree Hyperparameter Tuning Curve
(F1-score vs. Maximum Depth)

[Random Forest]

The tuning curve for the number of estimators shows that the training F1-score remains nearly perfect across all values, while the validation F1-score improves slightly and then stays relatively stable. This suggests that increasing the number of trees beyond a certain point does not significantly improve performance. The learning curve shows a large gap between training and validation F1-scores. The training score remains very high, while validation performance gradually improves as more data is added. This indicates some overfitting, but also shows that the model benefits from additional data and generalizes better with larger training sets. Overall, Random Forest achieves the strongest performance among the models but still exhibits some variance.

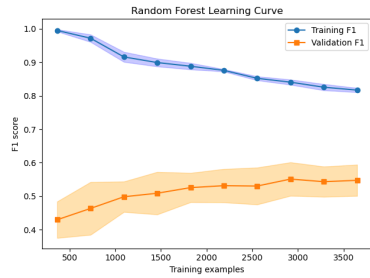


Fig. 4A. Random Forest Learning Curve
(F1-score vs. Training Set Size)

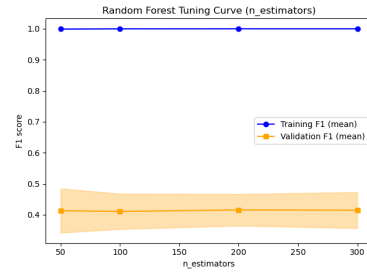


Fig. 4B. Random Forest Hyperparameter Tuning Curve
(F1-score vs. Number of Estimators)

Multiclass Classification of Breast Cancer Subtype

[Kernel SVM]

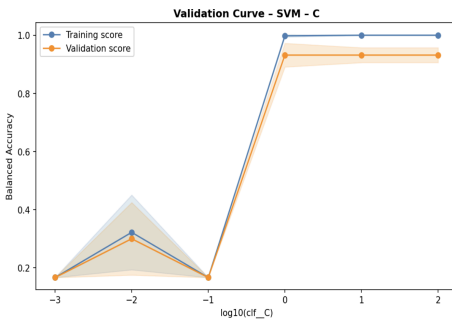


Fig. 5A. Validation curve showing the effect of regularization parameter C on Kernel SVM balanced accuracy across training and cross-validation sets

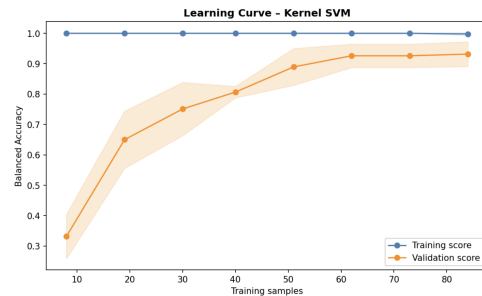


Fig. 5B. Learning curve for Kernel SVM showing training and validation balanced accuracy as a function of training set size

The validation curve for Kernel SVM shows that C values below 0.1 produced balanced accuracy near the random chance baseline of 0.17, confirming severe underfitting when the margin constraint is too tight in this high-dimensional space. At $C=1$, validation accuracy jumped sharply to approximately 0.93 and plateaued, confirming $C=1$ with $\gamma=\text{scale}$ as the optimal configuration. This sharp transition illustrates how sensitive kernel SVM is to regularization in high-dimensional genomic settings.

The Kernel SVM learning curve shows clear, persistent overfitting. The training score held at 1.0 across all training sizes, meaning the RBF kernel memorized every training fold completely via arbitrarily complex decision boundaries in the PCA-transformed feature space. The validation score rose from approximately 0.33 at 8 samples to 0.93 at full training size, but the training-to-validation gap never fully closed. The validation curve had not yet plateaued at the rightmost point, suggesting performance would improve with more data. The highest CV standard deviation of 0.0406 confirms this model is the most sensitive to fold composition of all four.

[Random Forest]

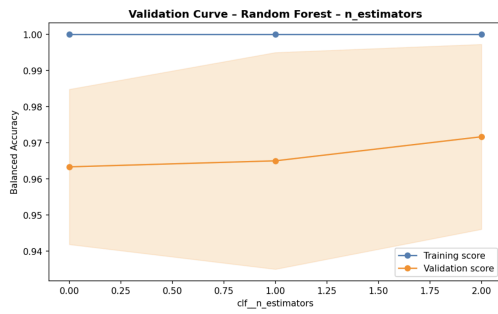


Fig. 6A. Validation curve showing the effect of $n_estimators$ on Random Forest balanced accuracy across training and cross-validation sets.

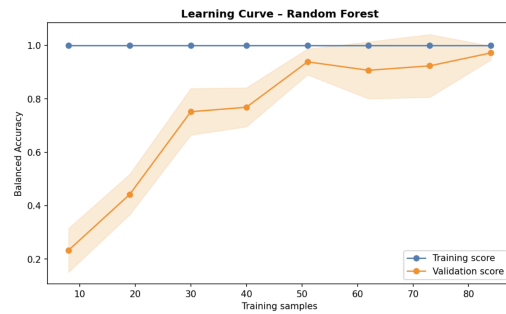


Fig. 6B. Learning curve for Random Forest showing training and validation balanced accuracy as a function of training set size.

The $n_estimators$ validation curve showed gradual improvement in validation balanced accuracy from 0.963 at 100 trees to 0.980 at 300 trees, while training score remained a perfect 1.0 throughout. Additional trees reduce variance through better ensemble averaging, and the diminishing gains beyond 300 trees confirm that 300 provides a good performance-to-computation tradeoff.

Random Forest showed the most textbook structural overfitting of all four models. Training score was a flat 1.0 at every training size from 8 to 85 samples, fully-grown unpruned trees partition any training set without error. The validation score started at 0.23 with 8 samples and climbed to approximately 0.97 at full training size, with a notably wide confidence band at intermediate sizes reflecting high fold-to-fold variance inherent to ensemble methods on small datasets. The validation curve largely plateaued around 70 samples, and the training-to-validation gap, while narrowing, remained visible at full size, a structural feature that would require max_depth pruning or a larger dataset to close.

[Logistic Regression]

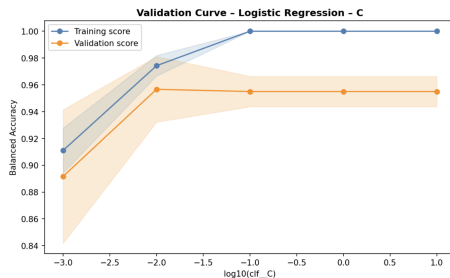


Fig. 7A. Validation curve showing the effect of regularization parameter C on Logistic Regression balanced accuracy across training and cross-validation sets

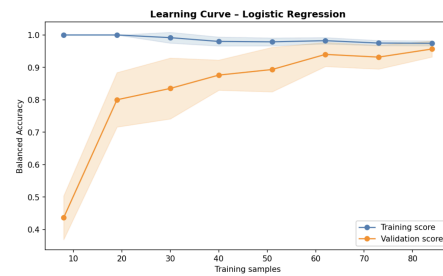


Fig. 7B. Learning curve for Logistic Regression showing training and validation balanced accuracy as a function of training set size.

At $C=0.001$, both training and validation scores hovered near 0.89, indicating the L2 penalty was too strong and the model was underfitting. As C increased to 0.01, both curves rose sharply to approximately 0.975 training and 0.956 validation, the point of best generalization and the selected optimal value. Beyond $C=0.01$, training continued climbing toward 1.0 while validation plateaued near 0.955, confirming that higher C values relax regularization too far and introduce overfitting. Validation performance remained stable from $C=0.01$ through $C=10$, confirming the model is robust once well-regularized.

The Logistic Regression learning curve was the smoothest of all four models and shows the strongest evidence of healthy generalization. Unlike Kernel SVM and Random Forest, the training score declined steadily from 1.0 at 8 samples to approximately 0.97 at full size, a positive sign that L2 regularization is forcing generalization rather than memorization. The validation score rose consistently from approximately 0.44 to 0.96, and the two curves were visibly converging by the final training size with a gap of only approximately 0.01. The tightest confidence band of all four models is consistent with the lowest CV standard deviation of 0.0244, confirming Logistic Regression at C=0.01 with L2 penalty as the most stable and generalizable model.

[Naive Bayes]

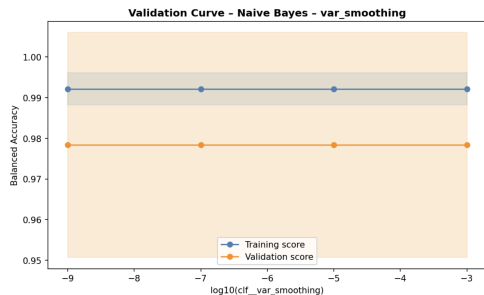


Fig. 8A. Validation curve showing the effect of var_smoothing on Naive Bayes balanced accuracy across training and cross-validation sets.

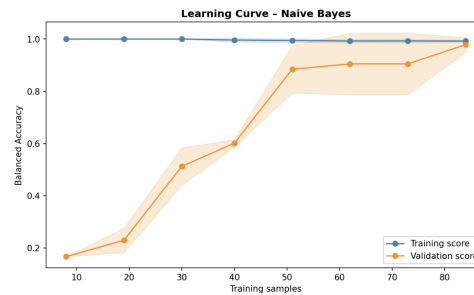
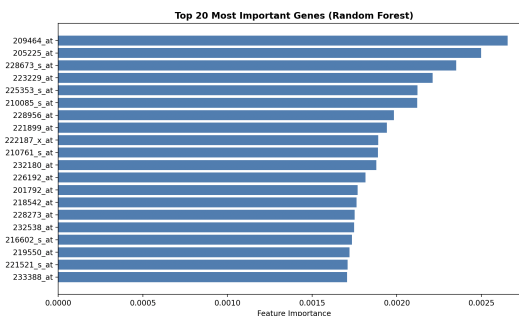


Fig. 8B. Learning curve for Naive Bayes showing training and validation balanced accuracy as a function of training set size.

The var_smoothing validation curve showed training at approximately 0.993 and validation at approximately 0.978, both essentially flat across all tested values from 1e-9 to 1e-3. Unlike the other three models, var_smoothing had virtually no effect on performance across four orders of magnitude, indicating that Naive Bayes behavior on this dataset is driven by SelectKBest feature quality rather than the smoothing hyperparameter. The very wide confidence band spanning 0.95 to above 1.0 confirms that fold composition, not hyperparameter choice, is the dominant source of performance variation.

Naive Bayes showed the most dramatic learning curve shape of all four models. With only 8 training samples, validation score started at approximately 0.17, the random chance baseline because estimating Gaussian distributions for 500 features across six classes from fewer than 9 samples produces statistically meaningless class-conditional estimates. Performance then made a near-vertical jump around 50 samples to approximately 0.89, before climbing to 0.98 at 85 samples. This step-function pattern reflects a critical mass threshold: below it, the probabilistic assumptions are unsupported; above it, they hold well enough for strong predictions. Despite this erratic behavior at small sizes, training and validation curves converged near 1.0 at full training size, giving Naive Bayes the smallest train-to-CV gap of any model, consistent with its feature independence assumption acting as an implicit regularizer.

[Additional Analysis — Random Forest (Multiclass)]



The cross-validation bar chart compares mean balanced accuracy and mean macro F1 across all four models, with error bars representing standard deviation across five folds. Naive Bayes and Random Forest achieved the highest mean CV balanced accuracy at 0.9783 and 0.9717 respectively, followed by Logistic Regression at 0.9567 and Kernel SVM at 0.9317. Critically, average performance alone does not identify the most deployable model, consistency matters equally. Logistic Regression's tightest standard deviation of 0.0244 makes it the most reliable across folds, while Kernel SVM's widest spread of 0.0406 signals sensitivity to training data that could hurt real-world performance. The close alignment between balanced accuracy and macro F1 across all models confirms both metrics tell a consistent story, validating that results reflect genuine learning rather than artifacts of class imbalance.

Results

[Binary Classification of Wine Quality]

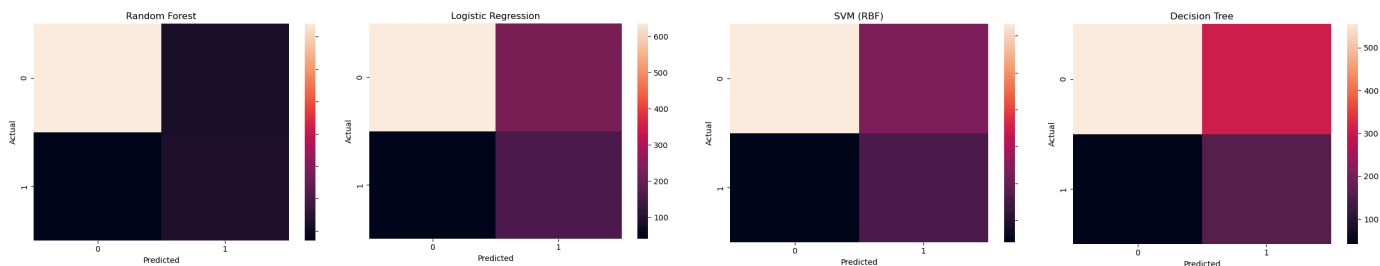
The optimal hyperparameters selected through grid search are as follows. Logistic Regression uses $C=1$ and $l1_ratio = 1$. The SVM (RBF) performs best with $C=1$ and $\gamma=0.1$. The Decision Tree uses $max_depth = 3$, $min_samples_leaf = 1$, and $min_samples_split = 2$. The Random Forest uses $n_estimators = 300$, $max_depth = 10$, and $max_features = "sqrt"$.

Wine Quality — Model Results Summary				
Accuracy F1 ROC-AUC across Training, Cross-Validation, and Test				
Model	Hyperparameters	Training Accuracy, F1, ROC-AUC	Cross-Val Accuracy, F1, ROC-AUC	Test Accuracy, F1, ROC-AUC
Random Forest	'max_depth': 10, 'max_features': 'sqrt', 'n_estimators': 300	Acc: 0.917 F1: 0.817 ROC-AUC 0.976	Acc: 0.810 F1: 0.547 ROC-AUC 0.844	Acc: 0.819 F1: 0.576 ROC-AUC 0.860
Logistic Regression	'model__C': 1, 'model__l1_ratio': 1	Acc: 0.736 F1: 0.534 ROC-AUC 0.829	Acc: 0.733 F1: 0.530 ROC-AUC 0.825	Acc: 0.748 F1: 0.546 ROC-AUC 0.837
SVM (RBF)	'model__C': 1, 'model__gamma': 0.1	Acc: 0.770 F1: 0.590 ROC-AUC 0.883	Acc: 0.746 F1: 0.547 ROC-AUC 0.841	Acc: 0.745 F1: 0.543 ROC-AUC 0.859
Decision Tree	'max_depth': 3, 'min_samples_leaf': 1, 'min_samples_split': 2	Acc: 0.701 F1: 0.515 ROC-AUC 0.809	Acc: 0.692 F1: 0.499 ROC-AUC 0.794	Acc: 0.672 F1: 0.478 ROC-AUC 0.797

Fig. 9. Wine Quality — Model Results Summary (Accuracy, F1-score, and ROC-AUC across Training, Cross-Validation, and Test Sets)

[Disclaimer: The information from this chart will be output in the terminal not as a chart this was merely to more easily visualize the information]

Random Forest achieves the best overall performance on the test set, with the highest test F1-score (0.576) and strong ROC-AUC (0.860), indicating effective modeling of nonlinear relationships. Logistic Regression performs consistently across all splits, showing stable generalization with only slightly lower F1-score. The SVM achieves similar performance to Logistic Regression but shows slightly higher variance and sensitivity to hyperparameters. In contrast, the Decision Tree performs the worst, with lower test scores and clear signs of overfitting. The confusion matrices show that all models perform well on the majority low/average-quality class but have more difficulty identifying high-quality wines. Random Forest provides the best balance, correctly identifying more high-quality wines while maintaining strong overall accuracy. Logistic Regression and SVM perform similarly but produce more false negatives. The Decision Tree shows the weakest performance, with more misclassifications and a stronger bias toward the majority class.



[Breast Cancer Gene Expression (Multiclass Classification)]

After hyperparameter tuning, the optimal parameters identified were: Kernel SVM with $C=1$ and $\gamma=\text{scale}$; Random Forest with $n_estimators=300$ and $\text{max_features}=\text{sqrt}$; Logistic Regression with $C=0.01$ and $\text{penalty}=\text{L2}$; and Naive Bayes with $\text{var_smoothing}=1e-5$ and $K=500$. The table below summarizes training, validation, and test performance for each model across both key metrics. Each cell reports Balanced Accuracy followed by Macro F1-Score:

Breast Cancer Gene Expression — Model Results Summary Balanced Accuracy Macro F1 · across Training, Cross-Validation, and Test							
Model	Hyperparameters	Training		Cross-Val		Test	
		Train Bal Acc	Train Macro F1	CV Bal Acc	CV Macro F1	Test Bal Acc	Test Macro F1
Kernel SVM	kernel: rbf C: 1 gamma: scale	0.9980	0.9982	0.9317	0.9312	0.8853	0.9017
Random Forest	n_estimators: 300 max_features: sqrt	1.0000	1.0000	0.9717	0.9702	0.9167	0.9357
Logistic Regression	C: 0.01 penalty: l2 SelectKBest k=300	0.9743	0.9744	0.9567	0.9542	1.0000	1.0000
Naive Bayes	var_smoothing: 1e-05 SelectKBest k=500	0.9922	0.9930	0.9783	0.9776	0.9687	0.9669

Fig. 11. Breast Cancer Gene Expression — Model Results Summary (Balanced Accuracy and Macro F1-score across Training, Cross-Validation, and Test Sets)

Logistic Regression achieved perfect test performance (1.000 balanced accuracy and 1.000 macro F1), making it the strongest model on the held-out set and the most effective for accurately classifying all six breast cancer subtypes. This is especially important given the project's goal of supporting clinical decision-making, where correctly identifying each subtype directly impacts treatment selection. Naive Bayes also performed strongly (0.9687 balanced accuracy, 0.9669 macro F1), showing reliable generalization despite the high-dimensional setting. Random Forest achieved high cross-validation performance (0.9717) but dropped to 0.9167 on test, indicating mild overfitting, while Kernel SVM had the lowest test performance (0.8853 balanced accuracy, 0.9017 macro F1) and the largest validation-to-test decline, reflecting sensitivity to data splits. Overall, all models significantly outperformed the random chance baseline (~ 0.17), confirming that gene expression data contains strong signals for distinguishing cancer subtypes. However, in this high-dimensional, low-sample setting, Logistic Regression's combination of perfect test performance and strong generalization makes it the most clinically reliable choice for accurately and consistently predicting breast cancer subtypes.

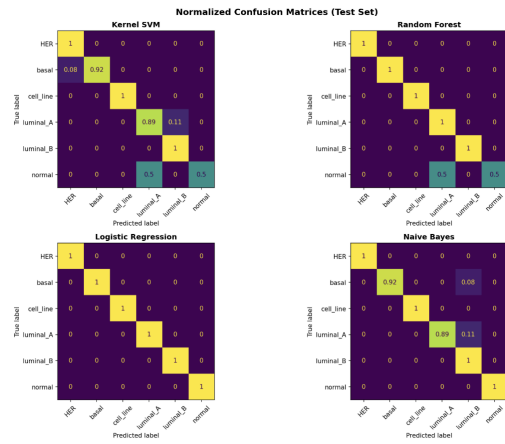


Fig. 12. Normalized confusion matrices for all four models evaluated on the held-out test set, showing per-class classification rates across the six breast cancer subtypes

The normalized confusion matrices revealed distinct per-class error patterns. Kernel SVM misclassified 8% of Basal samples as HER and 50% of Normal samples as Luminal A (1 out of 2 test samples). Random Forest similarly misclassified 50% of Normal as Luminal A but correctly classified all Basal samples. Naive Bayes correctly classified all Normal and Cell Line samples but misclassified 11% of Luminal A as Luminal B. Logistic Regression produced a perfect diagonal confusion matrix with zero misclassifications across all six classes.

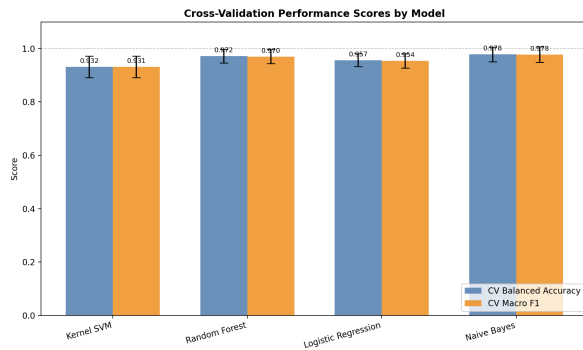


Fig. 13. Grouped bar chart comparing mean cross-validation balanced accuracy and macro F1 scores across all four models, with error bars representing standard deviation across the five folds.

The cross-validation bar chart compares mean balanced accuracy and macro F1 across all four models, with error bars representing standard deviation across five folds. Naive Bayes achieved the highest mean CV balanced accuracy (0.9783), followed by Random Forest (0.9717), Logistic Regression (0.9567), and Kernel SVM (0.9317). However, average performance alone does not determine the most reliable model. Logistic Regression had the lowest standard deviation (0.0244), making it the most consistent across folds, while Kernel SVM had the highest variability (0.0406), indicating sensitivity to data splits. Naive Bayes and Random Forest, although strong in average performance, showed slightly higher variability than Logistic Regression. This highlights a tradeoff between performance and stability: Naive Bayes offers the highest performance, while Logistic Regression provides the most reliable and consistent predictions. The close alignment between balanced accuracy and macro F1 across all models confirms that the results reflect genuine learning rather than artifacts of class imbalance.

Discussion

For binary classification, Random Forest performed best, achieving the highest test F1-score and strong ROC-AUC. This is expected for tabular data, as it captures nonlinear relationships and reduces variance through ensembling. In contrast, the Decision Tree performed worst, showing clear overfitting with lower validation and test performance. Logistic Regression and SVM (RBF) showed similar, moderate performance. Logistic Regression demonstrated stable generalization with low variance, while SVM captured more complex patterns

but was more sensitive to hyperparameters and data variation. Learning curves confirmed these trends, with Logistic Regression showing low variance, SVM moderate variance, and Decision Tree high variance. The confusion matrices revealed that all models performed well on the majority class but struggled to identify high-quality wines. Many errors were false negatives, reflecting both class imbalance and overlap near the 6–7 quality boundary.

For multiclass classification the cross-validation results reveal an important distinction between average performance and deployment reliability that is particularly consequential in a clinical context. While Random Forest and Naive Bayes led on mean CV balanced accuracy at 0.9800, Logistic Regression's tightest standard deviation of 0.0215 makes it the most consistent model across folds, meaning its observed performance is the most likely to reflect what would be seen on new patient samples. Kernel SVM's standard deviation of 0.0406, the widest of all four models, indicates that its performance is highly sensitive to which specific samples appear in each fold, a property that is especially problematic when the dataset contains only 7 Normal samples and 14 Cell Line samples. A misclassification of a Basal subtype as Luminal A, for example, could lead to a patient receiving hormone therapy when they require a more aggressive chemotherapy regimen. This reframes the model selection question: the best model is not necessarily the one with the highest mean score, but the one whose performance ceiling is high enough and whose variance is low enough to be trusted on any individual patient, regardless of how similar or different they are to the training distribution. By that standard, Logistic Regression's combination of a 0.9567 CV score, a 1.0 test score, and the tightest fold-to-fold variability makes it the most clinically defensible choice among the four models evaluated, even if its average cross-validation score does not lead the comparison.

Some overfitting remains due to the dataset's structure, with over 54,000 features and only 105 training samples. This extreme feature-to-sample ratio makes it easy for models to learn noise rather than generalizable patterns. While feature reduction helped, fully resolving overfitting would require a larger dataset, which is a common limitation in gene expression studies. The only fundamental solution would be a larger patient cohort, which is a known limitation of rare cancer subtype datasets in biomedical research. The feature reduction pipeline reduced the overfitting gap meaningfully but closing it entirely without more data is not feasible.

Overall, the two datasets highlight different modeling challenges. The wine dataset benefits from flexible ensemble models due to nonlinear feature interactions, while the breast cancer dataset favors simpler, well-regularized models due to extreme dimensionality and small sample size.

Conclusion

This analysis demonstrates the importance of model selection and proper evaluation when working with imbalanced tabular data. Among the models tested for binary classification, Random Forest achieved the best overall performance, with the highest F1-score and strong ROC-AUC, making it the most suitable model for predicting wine quality. Logistic Regression and SVM also performed well, offering stable and competitive results, while the Decision Tree showed weaker generalization due to overfitting. A key takeaway for this dataset is that class imbalance and overlapping feature distributions limit model performance, particularly in identifying high-quality wines. Even with class weighting and undersampling, many errors occurred near the decision boundary, highlighting the inherent difficulty of the task.

For the multiclass breast cancer dataset, model performance was driven more by stability than raw accuracy. Naive Bayes and Random Forest achieved the highest cross-validation scores (0.9783 and 0.9717), while Logistic Regression achieved perfect test performance (1.0). However, given the small test set, this result should be interpreted cautiously. Logistic Regression's lower variability across folds suggests it is the most reliable model overall, while Kernel SVM performed the worst due to overfitting in the high-dimensional, low-sample setting. All models performed well above the random baseline (~ 0.17), confirming that gene expression data contains strong predictive signals.

Comparing the two datasets, the results show that model effectiveness depends heavily on data characteristics. The wine dataset benefits from flexible ensemble models that capture nonlinear relationships, while the breast cancer dataset favors simpler, well-regularized models due to its extreme dimensionality and limited sample size. With more time and resources, several improvements could be explored. For the wine dataset, additional feature engineering and alternative imbalance handling techniques could improve detection of high-quality wines. For the breast cancer dataset, larger patient cohorts would help reduce overfitting and improve reliability. More advanced dimensionality reduction methods, such as UMAP, and ensemble approaches like stacking could further improve performance. Future work could also explore incorporating domain-specific features and graph-based models to better capture underlying relationships in the data.

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